

AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Title & Team Details

Project: AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Team: [Your Team Name]

Member 1: Pranjali Srivastava – AI/ML, NLP, RAG, Knowledge Graph, Predictive Maintenance

Member 2: [Partner Name] – Backend APIs, Database, Flask Integration, UI.

Problem Overview

Industrial organizations store maintenance manuals, SOPs, sensor logs, and reports across disconnected systems. Finding accurate information is slow and knowledge is lost when experienced personnel leave. Our platform unifies these sources into an AI-powered knowledge system.

Proposed Solution

The platform ingests PDFs, DOCX, Excel files, images, and NASA CMAPSS datasets. OCR extracts text from scanned documents, NLP extracts entities and relationships, embeddings are stored in FAISS, a Knowledge Graph stores structured relationships, and an LLM answers user questions using Retrieval-Augmented Generation (RAG).

System Architecture / Workflow

Document Upload → OCR & Text Extraction → Chunking → Embedding Generation → FAISS Vector Store → Entity & Relationship Extraction → Knowledge Graph → User Query → Semantic Retrieval → LLM Response → Predictive Maintenance Dashboard.

Technologies

Python, Flask, PyTorch, Hugging Face Transformers, Sentence Transformers, FAISS, spaCy, NetworkX, Tesseract OCR, Pandas, NumPy, Scikit-learn, LangChain-ready RAG pipeline, HTML/CSS/JavaScript.

Implementation & Results

Implemented features: document ingestion, OCR, entity extraction, relationship extraction, Knowledge Graph, semantic retrieval, predictive maintenance using NASA CMAPSS, recommendation engine, explainable AI, web dashboard. Include screenshots of dashboard, upload page, chat page, entity extraction, knowledge graph, and prediction page.

Performance Metrics

Entity extraction confidence up to 1.0 for industrial entities. Random Forest models trained for Remaining Useful Life prediction and failure classification. Report MAE, RMSE, R^2 , classification accuracy, and confusion matrix from your latest experiments.

Business Impact & Scalability

Applicable to manufacturing, power plants, oil & gas, aviation, and smart factories. Reduces document search time, preserves organizational knowledge, supports predictive maintenance, and improves operational efficiency. Future work includes graph databases (Neo4j), streaming IoT data, multi-modal LLMs, cloud deployment, and enterprise authentication.